جامعة خليفة
Khalifa University

ELECTRICAL SAFETY MANAGEMENT

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1. Purpose:

The purpose of this guideline is to provide the framework for controlling the risks associated when working with electricity to the University's workers, students and visitors.

This document is based on specific legislative requirements as contained in the Law No (9) of 2009 Amending Certain Provisions of Law No (2) of 1998 Concerning the Regulation of the Water and Electricity Sector in the Emirate of Abu Dhabi and the Electrical Wiring Regulations 2007.

In addition to legislative requirements, the general principles of risk management (identification, assessment, control, and review) should be applied to electrical hazards. This procedure outlines the controls specific to certain hazards concerned with electricity safety.

2. Scope:

This procedure outlines the University's electrical safety program, aimed at reducing the risk of exposure to electrical hazards for staff, students, visitors and contractors. The provisions established within this document are applicable to all areas within the University in which work, teaching or research is undertaken.

3. Responsibilities:

3.1 Deans, Directors, Heads of Departments: are to ensure that:

- The Electrical Safety Management Procedure is implemented within the local areas they are responsible for. Implementation includes the appointment of testing personnel to perform the testing requirements.

3.2 Area Supervisors/Coordinators: Responsible for

- Ensure that testing and tagging is being carried out in their respective area.
- Ensure that any faulty equipment is dealt with in the appropriate manner
- Act as a point of contact for the electrical testers for the purpose of induction of individual laboratories or workshops etc.
- Provide instructions to the electrical tester regarding specialist equipment that may be in the area.
- Ensure that electrical testers are appropriately paid for their work.
- Those working in an area are aware of the tester's impending visit.
- Ensure that a list of areas to be inspected is provided to the electrical tester.

3.3 EHS Manager: Responsible for

- Overall responsibility of the implementation, monitoring and effectiveness of this procedure.
- Communicate and ensure the awareness this procedure to all levels of KU.
- Review the lessons learned from the implementation of this procedure.
- Ensure that all training requirements related to this procedure is identified through coordination with Section Managers.

3.4 Testers of Electrical Equipment: are to ensure that:

- Prior to entering an area they organize with the Local Area Coordinator an appropriate time for induction into the area and for the testing of equipment
- All electrical equipment requiring testing within the work area under their control is inspected, tested, tagged and recorded in accordance with this policy and procedure
- Any faulty electrical equipment is tagged appropriately and removed from service until it satisfies the testing requirements.
- If the test fails, they must attach a KU **"Out Of Service Tag"** on lead and notify the local area coordinator.
- Satellite Campuses' electrical items requiring testing will be tested and tagged according to this guideline by a third party contractor. This may include equipment from the main campus that has been transported to a satellite campus.

3.5 Workers: are to ensure:

- Any electrical equipment that has been tagged with a Danger/Out of Service Tag must not be used and must be removed from service until an appropriately licensed person has completed repairs
- Items requiring testing are available when the tagger is scheduled
- Suspect items are checked by a qualified person before use.

In addition, all users of electrical equipment should:

- Visually inspect all electrical equipment prior to use to ensure:
  - There is no obvious external damage, particularly to plugs, sockets, cords or other connectors.
  - The flexible supply cord is free of damage, anchored correctly and no inner core insulation is visible.
  - Covers and guards are correctly secured.
  - Ventilation inlets or exhausts are not obstructed.

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- Ensure the appropriate area supervisor is notified of any faults and equipment that is out of service for maintenance or repair.
- Use the correct appliance for the specific task.
- Ensure that electrical appliances are dry and clean.
- Ensure that an appropriate power board is used where more than one appliance is required to be connected to a single GPO/ extension cord.
- Not use double adaptors or “piggyback” plugs
- Not withdraw a plug from a socket by pulling the cable.

3.6 EHS inspectors: Responsible for

- Conducting regular audits on work areas and electrical equipment.
- Close monitoring & performance of this procedure.
- Assist all supervisors and employees concerned with the management of electrical hazards.
- Report all incidents and near misses.

4. Abbreviation / Definitions:

Electrical Competent Person: A Competent Electrical Person is an Appointed Person who has demonstrated that he has sufficient knowledge, training and experience and judged as competent to avoid danger from live electrical equipment and to carry out duties under the authority of the Electrical Safety Rules.

Electrical Appliance: A device or apparatus that is connected to the electrical power supply through a general-purpose outlet in the form of a plug-in.

Electrical Installation: A system that supplies electricity to a building, including the switchboards, distribution boards, fixed wiring and socket outlets.

Extension Lead: An assembly of a plug intended for connection to a mains outlet socket, a sheathed flexible cord and a cord extension socket.

Electrical/Electronic Waste: Discarded, surplus, obsolete, or broken electrical or electronic devices.

Electrical Work: Work involving:

- Connecting electricity supply wiring to electrical equipment or disconnecting electricity supply wiring from electrical equipment.

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- Installing, removing, adding, testing, replacing, repairing, altering or maintaining electrical equipment or an electrical installation

Fixed Equipment: Equipment that is fastened to a support, secured in position or otherwise, due to its size and mass located in a specific location.

Flexible Supply Lead: A flexible cable or cord, which supplies voltage to electrical equipment. It has one end connected to a plug with pins designed to engage with a socket outlet (GPO), and the other end either:

- Connected to terminals within the equipment; or
- Fitted with a connector designed to engage with an appliance inlet fitted to the equipment.

General Purpose Outlet (GPO): A three-pin wall socket commonly used to deliver 240 volts.

Hostile Operating Environment: An operating environment at a place of work where an item of electrical equipment is, in its normal use, subject to operating conditions likely to result in damage to the item of equipment. This includes an operating environment that may:

- Cause mechanical damage to the item of equipment, or
- Expose the item of equipment to moisture, heat, vibration, corrosive substances or dust that is likely to result in damage to the item of equipment.

For example: workshops, research lab, kitchen, kitchenettes, tearooms, toilets, amenities, shower areas, general campus areas, student housing and field stations.

Non-hostile operating environment: A workplace that is dry, clean, well-organized and free of operating conditions that are likely to result in damage to electrical equipment or the flexible supply cord. For example: an office, retail shop, computer lab, lecture theaters, classrooms, etc.

Personal Equipment: Any privately owned electrical equipment of the plug-in type to be used at the University. Examples of personal equipment include (but is not limited to) radios, electric heaters and fans, sandwich makers, coffee makers, and mobile phones.

Personal, leased or hired electrical equipment: Electrical equipment not purchased or owned by KU but used in a KU workplace. Such equipment may include: but is not limited to, laptop computers; heaters; fans; handheld power tools; audio equipment; kitchen appliances; and electronic test equipment.

Portable Appliance Tester (PAT): Is an electronic instrument that automatic test equipment plugged into it. The results it indicates required no technical interpretation.

Power Board: A device having a single plug intended for connection to a mains outlet socket, a sheathed flexible cord and an assembly of one or more outlet sockets.

Permit-To-Work System: The system that allows central control and ongoing monitoring of higher risk activities to ensure that activities are:

- Authorized;
- Carried out by qualified personnel using appropriate safety precautions; and
- Those activities with potentially hazardous interactions do not take place at the same time.

Activities included in the system should include:

- Hot works;
- Confined spaces;
- Electrical work;
- Working at heights;
- Roof access;
- Excavation and trenching; and
- Energy isolation.

RCD: A Residual Current Device is a mechanical switching device intended to isolate the circuit when the current imbalance attains the rated operating leakage current value of the device.

RCD's are classified according to their rated residual current as follows:

- Type I: <10mA
- Type II: >10mA<30mA
- Type III: >30mA < 300mA without selective tripping time delay
- Type IV: >30mA < 300mA with selective tripping time delay

Only Type I and Type II are suitable for personal protection in the event of a current flow to earth.

Risk Management: The process of implementing decisions about accepting or altering risks.

Stationary Equipment: Equipment having a mass exceeding 18 kg and not provided with carrying handle(s).

Tag: A durable, non-reusable, non-metallic label that may be color coded to identify the period in which testing was performed.

Tag-out: "Tag-out" device is a prominent warning device, such as a tag with a means of attachment, which can be securely fastened to indicate that the energy-isolating device and the equipment being controlled may not be operated until the tag-out device is removed.

The placement of a tag-out device on an energy-isolating device indicates that the energy-isolating device and the equipment being controlled may not be operated until the tag-out device is removed.

Voltage: The difference in electrical potential between two points in a circuit, normally existing between conductors and between conductors and earth expressed in volts.

Extra-Low Voltage: Voltage not exceeding 50V a.c. or 120 V ripples free d.c.

Low Voltage: Voltage exceeding extra-low voltage, but not exceeding 1000V a.c. or 1500V.

High Voltage: exceeding low voltage.

Worker: Any person who carries out work for the University, which includes:

- Employees
- Trainees
- Volunteers and affiliates including visiting and honorary fellows
- Outworkers
- Apprentices
- Work experience students
- Contractors or sub-contractors
- Employees of a contractor or sub-contractor
- Employees of a labor hire company assigned to work for the University.

5. Procedure:

5.1 Risk Management:

Hazards associated with electrical work can arise from a range of aspects associated with the type of work being undertaken. The requirements set

out in the KU Risk Management Procedure are applied for the identification of hazards, assessment and control of risk when performing electrical work. A systematic risk management approach must be applied to eliminate or control the risk of electrical hazards. A range of control measures can be implemented including:

- Routine visual checks;
- Regular inspection;
- Maintenance;
- Repair;
- Replacement;
- Use of residual current devices (RCDs);
- And, where warranted, testing of identified electrical equipment.

5.2 Implementation:

5.2.1 Type of Electrical Appliances to be Tested:

The University must ensure that electrical equipment is regularly inspected and tested by a competent person if the electrical equipment is:

- Supplied with electricity through an electrical socket outlet ('plug in' equipment), and
- Used in an environment in which its normal use exposes the equipment to operating conditions that are likely to result in damage to the equipment or a reduction in its expected life span.

This includes conditions that involve exposing the electrical equipment to moisture, heat, vibration, mechanical damage, corrosive chemicals or dust. Examples may include work environments such as construction and demolition sites, however testing may also be required for items in office environments if the equipment is exposed to the conditions listed above. Table 1 outlines the type of equipment and operating environments where inspection and testing is required.

Operating Environment	Examples of Electrical Equipment Requiring Inspection and Testing	
Workshop	• Hand held power tools • Laboratory equipment • Portable electrical equipment	• Electronic plant • Extension cords • Power boards
Cleaning activity	• Floor polishers	• Vacuum cleaners
Commercial kitchen	• Commercial kitchen appliances	• Portable electrical equipment

Common Teaching Area	• Laptop computers • Overhead projectors	• Portable electronic whiteboards
Construction site	• Extension cords • Hand held power tools	• Portable electrical equipment • Power boards
Laboratory	• Flow Cytometer • Plate Reader • Gamma Counter • Fast Liquid Protein • Chromatography (FLPC)	• Hot Plates • Stirrers • Small Drying Oven • Hair Dryer
Lecture theatre	• AV equipment • Smart Boards	• Overhead projectors
Theater/Studio	• Lanterns • Lighting Poles	• Stage lights
Workshop	• Electronic plant • Extension cords	• Power boards • Powered hand tools

Table 1: Examples of electrical equipment requiring inspection and testing

The categories in the following table 2 list plug-in type electrical equipment that is commonly used in higher-risk operating environments and does require regular inspection and testing, and provides examples of the types of electrical equipment.

Category of equipment	**Examples of electrical equipment to be tested**
Hand held electrical equipment	• Hand held power tools • Hairdryers • Kitchen appliances • Laboratory equipment
Portable electrical equipment moved while in operation	• Floor polishers • Vacuum cleaners • Portable lighting equipment
Electrical equipment that is moved between operations in such a manner that could damage the flexible supply lead	• Portable electronic whiteboards, overhead projectors • Laptop computers • Electrical plant used in factory type environments • Welding machines • Extension cords • Power boards
Electrical equipment that is	• Electrical equipment used in wet or dusty

used in a higher-risk operating environment where damage to the equipment or the electricity supply to that equipment could occur such as in wet or dusty conditions.	areas • Electrical equipment used outdoors, in kitchens, • Laboratories (chemical damage) • Certain factory-type environments

Table 2: list plug-in type electrical equipment that is commonly used in higher-risk operating environments

In some situations, electrical equipment may not require inspection and testing as it does not present a risk to the operator due to:

- The permanent nature of its location
- The way the equipment is used
- The working environment in which the equipment is used.

Examples of equipment that do require inspection and testing and recommended testing intervals can be found in Annexure 1: Inspection and Testing of Electrical Equipment Intervals.

Examples of equipment where electrical testing and tagging is not normally required as they are usually located in a non-hostile operating environment include:

- Desktop computers and monitors
- Scanners
- Photocopiers
- Lamps
- Radios
- Fridges
- Microwaves
- Freezers.

If there is doubt over the operating conditions of the equipment a risk assessment should be completed as outlined in section 5.2.2 to determine whether an inspection is required as a risk control measure.

5.2.2 Risk Assessment:

If a risk assessment is carried out as per the requirements of section 5.1 it should be done so in accordance with the KU Risk Management Procedure. This process should involve a competent person. The results of the completed risk assessment will determine whether the electrical equipment requires testing and tagging.

A risk assessment should be conducted for any equipment that is not listed in this procedure as requiring inspection and testing to determine whether it is appropriate or not to do so. The risk assessment should take into account the conditions the equipment is exposed to, the safe operation of the equipment and previous incidents.

5.2.3 Equipment:

5.2.3.1 Previously Tagged Equipment:

A competent person must remove the old or out of date tag when equipment has been previously tested and tagged and, after the completion of a risk assessment, is deemed to no longer require on-going testing and tagging.

5.2.3.2 New Equipment:

Where new equipment is introduced to the workplace, the supplier is responsible for providing a safe product and thus electrical testing and tagging is generally not required. Where the supplier has deemed the equipment to require electrical testing as part of the commission process, the University will perform the electrical testing.

If any new equipment is deemed to require ongoing testing as it should be tagged to determine the initial test date and subsequent testing frequency thereafter.

5.2.3.3 Second Hand Equipment:

Where new equipment is introduced to the workplace, the person bringing in the product must ensure that it is tested and tagged. The retesting period should be highlighted on the electrical tag.

5.2.3.4 Imported Equipment:

All electrical items that have been imported from overseas must be safe to use in the workplace. Any electrical item that is located at a KU work place must conform to AD EHSMS COP15.0. Any equipment that is purchased from overseas must be inspected by a qualified electrician before it is placed in the workplace and allowed to be operated. If the inspection process identifies that the equipment does not conform to AD EHSMS COP15.0 then it must be serviced and modified so that it does. This inspection process should be demonstrated by completing an Imported Electrical Equipment Inspection Checklist.

5.2.3.5 Electrical Equipment That Is Not Able To Be Tested By A PAT:

Electrical equipment requiring testing that is not able to be tested with a PAT, for example three phase equipment or voltages other than 240v, should be conducted by an electrician.

5.2.3.6 Personal Equipment:

Electrical inspection and testing of personal equipment is required to be tested as per Annexure 1: Inspection and Testing of Electrical Equipment Intervals. Use of personal equipment at the University is to be approved by the Facilities Management Department (FMD). Personal equipment which is not approved by the FMD is prohibited from use at the University.

5.2.4 Electrical Testing and Tagging Flowchart:

A flowchart for those undertaking the testing and tagging of electrical equipment can be seen in Figure 1: Process for testing and tagging electrical equipment. This is to be used in conjunction with the Electrical Testing and Tagging Procedure.

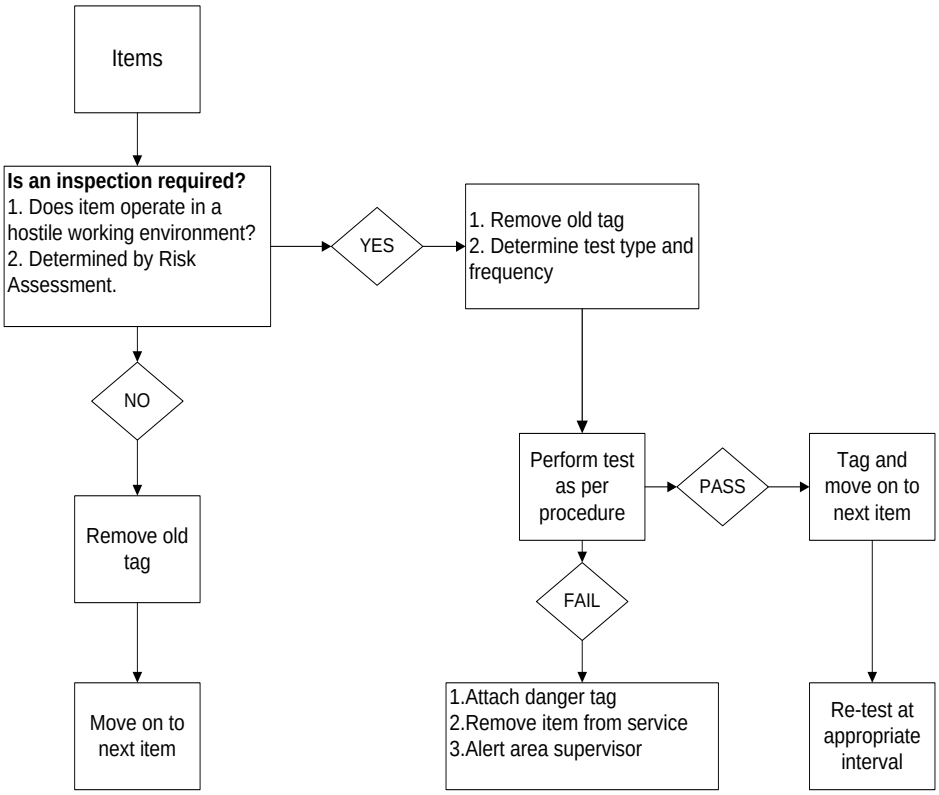

Figure 1: Process for testing and tagging electrical equipment

5.2.5 Testing Intervals of Electrical Appliances:

Intervals for testing and inspection of electrical appliances shall not exceed those specified in Annexure 1: Inspection and Testing of Electrical Equipment Intervals.

Factors considered in determining appropriate inspection and test intervals include the mobility of equipment, flexion of the cord and the environment where the equipment is used to evaluate the risk to health and safety.

The intervals in Appendix 1 are the minimum requirements for testing at the University. Additional tests may be undertaken to determine the safety of electrical equipment as necessary. This is at the discretion of the relevant supervisor in consultation with staff members.

5.2.6 Training Requirements for Testing Personnel:

Inspection and testing of electrical equipment should be carried out by a competent person. For this reason electrical testing will only be carried out by a qualified electrical or electronic tradesperson or any person trained to use a portable appliance tester (PAT).

5.2.7 Inspection and Test Procedure:

Inspection and testing of electrical appliances are to be in accordance with the University's Electrical Inspection and Testing Operating Procedure.

5.2.8 Compliant Electrical Appliances:

Electrical appliances which are compliant with the inspection and testing procedure are to be fitted with an appropriate tag. The tag must include the asset identification number, the identity of the person carrying out the testing, the date tested and the date due for retest.

5.2.9 Noncompliant Electrical Appliances:

Electrical appliances which are non-compliant with the inspection and testing procedure are to be fitted with a danger tag warning persons of a potential hazard. The item is to be withdrawn from service (if possible), the area supervisor is to be informed and the item is to be isolated to ensure that it cannot be used.

Removal of the plug may be appropriate to ensure that equipment cannot be used. This should only be undertaken by a competent person and when the equipment is unplugged from the power point and is safe to do so.

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5.2.10 Equipment Requiring Repair:

Where the equipment is required to be repaired, and does not comply with the CE standard, prior to work being conducted it should be verified that the equipment has an earth.

Repair of equipment shall only be completed by authorized repair agents or suitably qualified electrical personnel. Personal equipment is to be repaired at the owner's expense. The equipment must be re-tested in accordance with Standard for compliance and then tagged according to this guideline prior to re-introduction to service.

Equipment being returned to service should be accompanied by a document stating the item is safe for use when returned to service. This document should include details such as date returned to service, name, signature and company of the person who is returning the item to service.

5.2.11 Record Keeping:

Records of inspection and testing of electrical equipment are required. The University recommends the use of PAT testing equipment compatible with electric testing records database to aid in the maintenance of testing records.

Records of maintenance should be kept throughout the working life of the equipment. Copies are to be retained by the department.

Records are to include at minimum:

- The date of inspection and test
- Clear identification of the equipment tested
- The results of the test (whether the equipment passed or failed)
- The identification of the person carrying out the testing and
- The date retest is due.

5.3 Power Boards and Extension Sets:

Power boards used at the university must comply with the Electricity Wiring Regulations 2007. In essence, power boards are to have the following minimum features:

- Current overload protection, and
- Reset button.

Power boards with long leads (i.e. over 1.8m) are to have individual switches on the power board. In office environments power boards must be

located in an area which does not cause damage to the board or provide a tripping hazard. In all other locations they must be securely mounted clear of the workbench or floor. Power boards are not to be overloaded when in use, i.e. piggy back one board onto another.

Power boards that are not compliant with this guideline are to be removed from service. Due to the increased risk caused by double adaptors and 'piggyback' plugs, they are not to be used under any circumstances within the university.

Extension leads should be compliant to the Electricity Wiring Regulations 2007. When in use, extension leads are to be fully extended, not covered with mats, and not placed where they could be a tripping hazard (e.g. across aisles, corridors or other trafficable areas).

Power boards and extension leads are to be inspected and tested depending on their environment and flexion as per Table 1: Examples of electrical equipment requiring inspection and testing.

5.4 Residual Current Devices (RCD):

Any electrical risk associated with the supply of electricity to 'plug in' electrical devices must be minimized by the use of an appropriate RCD in certain higher-risk workplaces. If electricity is supplied to the equipment requiring an RCD through a socket outlet not exceeding 20 amps the RCD must have a tripping current that does not exceed 30 milliamps. However, this does not apply if the supply of electricity to the electrical equipment:

- Does not exceed 50 volts alternating current, or
- Is direct, or
- Is provided through an isolating transformer that provides at least an equivalent level of protection, or
- Is provided from a non-earth socket supplied by an isolated winding portable generator that provides at least an equivalent level of protection.

A RCD is required to be incorporated before or as part of the electrical outlet socket. Those areas using electrical handheld equipment, moveable equipment or equipment used in higher risk working environments must ensure either fixed or portable safety switches are used. RCDs offer a supplementary means of protection against electrocution.

Correct selection of the type of earth leakage protection is also important to avoid an unacceptable level of circuit tripping by the devices. The minimum

requirement for an RCD is a tripping current which does not exceed 30 milliamps when electricity is supplied not exceeding through a socket outlet 20 amps.

Testing and inspection of RCDs are to be completed in accordance with complying with BS EN 61008 and BS 4293. The intervals for testing of RCDs are set out in Annexure 2: KU Inspection and Testing of Residual Current Devices. Records must be kept of RCD testing.

5.4.1 RCD - Facilities Management RCD roll out strategy:

To meet RCD compliance, FM is currently rolling out an RCD upgrade strategy. The program has taken the following steps:

- Risk analysis undertaken to identify hostile environments.
- Identification of distribution boards servicing hostile areas but lacking RCDs.
- Engineering level control upgrade by retro fitting of RCDs to necessary boards.

5.4.2 Local area risk management – Portable RCD's:

During the FM roll out program, local areas containing hostile environments within their operations are still required to adequately control the risk of electrical shock. This can be achieved through the normal electrical safety / maintenance program. However, in addition portable RCD protection is also required for portable electrical equipment used in hostile environments. Portable RCDs must be able to meet the technical requirements of the task at hand (i.e. nominal voltage, tripping current, switching current etc.). Local areas can check technical requirements by consulting the following:

- Standards and Codes of Practice listed in the reference section of this procedure;
- Contact the listed procurement supplier;
- Consult a licensed electrician.

How can I get assistance with this?

- Contact FM Assist 8888 to find out whether the electrical supply board to your hostile environment is RCD-protected. If it is, no further action required.
- Ensure that all your equipment is electrically tested and tagged.
- Fit portable RCD protection on all electrical appliances in the hostile environment. Do this by purchasing a portable RCD, see the options below:

Examples of non-portable (fixed) protection (i.e. Facilities responsibility)

Example of Non-portable (fixed) RCD that can be fitted to a switchboard	Non-portable (fixed) SRCD incorporated within the powerpoint

Examples of portable protection

(Local area responsibility after identifying equipment used in hostile environments)

RCD attached to power cord	
Plug in RCD	
RCD protected portable outlet	

What equipment needs RCD protection?
Depending on the type of appliance, installation and environment, RCD protection may be provided by either portable or fixed RCDs.

Hand Held Electrical Plant:
- Power tools
- Hair dryers
- Electrical knives

Electrical Plant Which Is Moved During Operation:
- Vacuum cleaners
- Floor polishers
- Extension cords/ Power boards
- Portable lighting

Plant is moved between Operation Where Damage to Plant or the Supply Cord Could Occur:
- Electrical welders
- Portable bench saws
- Audio visual equipment
- Extension cords/ power boards

Where Electrical Safety could be affected by the Operating Environment:
- Appliances used in wet areas such as kettles and other kitchen appliances
- Any equipment is used in an environment where it is exposed moisture, heat, vibration, mechanical damage, corrosive chemicals or dust.

Equipment that does not need RCD protection
- Extra low voltage equipment (less than 50V Alternating Current);
- Equipment supplied by direct current systems (less than 120V ripple free Direct Current);
- Equipment supplied from an isolated winding from an unearthed generator that provides an equivalent level of protection;
- Equipment supplied from an isolating transformer that provides an equivalent level of protection;
- Specialized scientific equipment where the use of an RCD may compromise the operation of the equipment or safety of a patient. (However, steps should be taken to ensure a high level of safety

is maintained such as a more frequent and extensive testing program).

5.4.3 Testing of RCDs used in a workplace must be tested:
Regularly by a competent person to ensure the devices are working effectively. A record of testing (other than daily testing) must be kept until the device is next tested or disposed of.

There are two types of test required for RCDs

- The manual push button (Trip) test.
- The leakage to Earth (10mA or 30mA leakage) operating time test: using an electrically isolated RCD test instrument
  - The **Manual Push Button (Trip) Test** can be performed by the user to determine the RCDs tripping function and approximate tripping time.
  - The **Leakage to Earth operating time test** is usually performed by an electrician and measures how long the RCD takes to trip.

5.5 Electrical Work:
The University must take all reasonable steps to ensure that electrical work that is required to be undertaken by a licensed or registered electrical worker is undertaken by a worker that meets the relevant licensing or registration requirements. Specific requirements for certain types of electrical work have been outlined below.

5.5.1 Access to Electrical Switchboards:
Access to University switchboards and the resetting of circuit breakers is restricted to the Facilities Management Department (FMD) staff and approved contractors only. Reports of trips circuits and requests for resetting tripped circuits must be directed to FMD ext. 8888 or in an emergency to KU ext. 8200.

5.5.2 De-Energized Electrical Work:
Electrical work is not to be carried out on electrical equipment while the equipment is energized. Before electrical work is carried out on electrical equipment the equipment must be tested by a competent person to determine whether or not it is energized.

When electrical work is being performed on electrical equipment, each exposed part must be treated as energized until it is isolated and

determined not to be energized. De-energization needs to be determined by a test which is safe and effective, thereby incorporating the principle of **'TEST FOR DEAD BEFORE YOU TOUCH'**.

The worker carrying out the testing must understand the testing procedures and be competent in the use of the testing method. Panel voltmeters should not be the only method of testing to be used to determine whether an electrical part is de-energized.

Additionally each high-voltage exposed part of the equipment must be earthed after being de-energized.

Anyone carrying out electrical work for the University must ensure that electrical equipment that has been de-energized to allow for electrical work to be carried out cannot be inadvertently re-energized.

A safe work procedure applying the **'TEST FOR DEAD BEFORE YOU TOUCH'** should be completed within any local area required to perform electrical work. Where safe work procedures have been developed for electrical work or energized electrical work, that work must be carried out in accordance with the safe work procedure.

5.5.3 Low Voltage Isolation and Access:

Isolating electrical equipment or circuits from all relevant sources of electricity supply can involve the use of opening switches, removing fuses or links, opening circuit breakers or removing circuit connections. The following steps should be taken when organizing low voltage isolation:

- Consultation
- Isolation
- Securing the isolation
- Tagging
- Testing
- Retesting as necessary

Local isolation procedures can help to ensure the above steps are catered for during the working process. Effective isolation can be achieved by applying lock off processes or tagging systems. All reasonable steps must be taken to ensure that restoring electricity supply following isolation does not pose risks to health and safety at the workplace. If work is left unfinished then the workplace must be left in a safe state that will not put anybody's health and safety at risk.

5.5.4 Energized Electrical Work:

Work must not be carried out on energized electrical equipment. When work is carried out on energized electrical equipment the following must be conducted:

- A risk assessment
- Clear the work area and the point at which the electrical equipment can be disconnected or isolated from its electricity supply of obstructions
- Clearly mark or label the point at which the electrical equipment can be disconnected or isolated from its electricity supply
- Make sure that the point at which the electrical equipment can be disconnected or isolated from its electricity supply is capable of being operated quickly.

Any work carried out on energized electrical equipment by a competent person must take into account the following:

- The competent person has tools, testing equipment and PPE that are suitable for the work, have been properly tested and are maintained in good working order
- Work is to be carried out in accordance with a safe work method statement prepared for the work, and
- Work is to be completed with a safety observer who can implement emergency management procedures, rescue the person who is carrying out the work and has been assessed as competent in the last 12 months in resuscitating a person unless otherwise specified as not required within a completed risk assessment.

5.5.5 Working Near Energized Electrical Parts:

Electrical work can pose a risk to direct or indirect contact with exposed energized parts, for example installing of testing circuits on a switchboard adjacent to exposed live electrical parts. At times, the risks associated with undertaking this type of work can be equivalent to those associated with live electrical work.

Hazards associated with working near energize electrical parts need to be eliminated or controlled in accordance with the KU Risk Management Procedure to ensure the health and safety of those performing the work. Careful planning and preparation needs to be undertaken to ensure the safety of workers, students, contractors and visitors.

5.5.6 Tools and Equipment:

Hazards associated with the use of tools and equipment need to be eliminated or controlled in accordance with the KU Risk Management Procedure to ensure the health and safety of those performing electrical work. Tools and equipment might include ladders, scaffolds, insulating barriers, insulating mats and test instruments.

6. Reporting of Electrical Incidents:

All electrical incidents with the potential to cause serious injury are required to be reported to the Sector Regulatory Authority (SRA). Hence, any injury or incident involving electricity in the University must be reported to the EHS Section immediately or as soon as possible after the event. This can be achieved by telephoning the Section during normal business hours or, if after hours, telephoning Emergency. As soon as practicable, a Hazard and or Incident Report form is to be completed via the University's online reporting system, Safety Webpage.

On receipt of a Hazard and or Incident Report form the EHS Section will investigate immediately to ensure the appropriate corrective actions have been taken to prevent further injury.

It is also important that any person, who has suffered an electric shock, seeks medical treatment as soon as possible after the event, as effects from electricity may have a delayed effect.

Whenever an electrical incident has occurred, Facilities Management Department electrical personnel are to be contacted to inspect the electrical outlet and appliance. Any equipment that causes an electric shock or is faulty is to be inspected and tested prior to re-use.

7. Program Evaluation:

In order to ensure that this procedure continue to be effective and applicable to the University, it will be reviewed at least every three years by the EHS Section in consultation with the EHS Steering Committee. Conditions which might warrant a review of the guidelines on a more frequent basis would include:

- Reported hazards or injuries.
- Nonconforming systems.
- EHS Steering Committee concern.

Following the completion of any review, the procedure will be revised /updated in order to correct any deficiencies. These changes will be communicated via the EHS Steering Committee.

8. Reference Standards:

8.1 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – Elements 02 – Risk Management.

8.2 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – Elements 03 – Management of Contractors.

8.3 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – Elements 06 – Emergency Management.

8.4 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – CoP-15 Electrical Safety.

8.5 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – CoP-35 Portable Power Tools.

8.6 ISO 45001:2018 Occupational health and safety management systems – Requirements.

8.7 ISO14001:2015 Environmental Management System – Requirements.

8.8 The Electricity Wiring Regulations 2007.

8.9 Requirements for Electrical Installations (BS 7671: 2001). The Institution of Engineering and Technology (formerly the Institution of Electrical Engineers).

8.10 ADWEA Wiring Rules and Regulations for LV Installations 2003, Abu Dhabi Water and Electricity Authority.

8.11 AS 3760:2010 Australian and New Zealand Wiring Rules – AS/ANZ 3000:2007.

8.12 AS/NZS 3760:2010. In-service safety inspection and testing of electrical equipment.

8.13 AS/NZS 3190:2011 Approval and test specification – Residual current devices (current-operated earth-leakage devices).

8.14 AS/NZS 3000:2007 Electrical installations.

8.15 BS 7671: Requirements for Electrical Installations. IEE Wiring Regulations Seventeenth Edition.

8.16 BS EN 55011: Limits and methods of measurement of radio disturbance characteristics of industrial, scientific and medical (ISM) radio-frequency equipment.

8.17 BS EN 60335-1: Specification for safety of household and similar electrical appliances. General requirements.

8.18 BS EN 60335-2-90: Specification for safety of household and similar electrical appliances.

8.19 BS EN 60947: Specification for low-voltage switchgear and control gear.

8.20 BS EN 61008-1: Residual current operated circuit breakers without integral overcurrent protection for household and similar use (RCCBs). General rules.

8.21 BS EN 61009-1: Residual current operated circuit-breakers with integral overcurrent protection for household and similar uses (RCBO's). General rules

9. Annexure:

7.1 Annexure 1 – Inspection and Testing of Electrical Equipment Intervals.

7.2 Annexure 2 – KU Inspection and Testing of Residual Current Devices.

10. Records / Documents / Formats:

10.1 Inspection Testing of Electrical Equipment Intervals Form No.: KU – FD – EHS – FOR – 121.

10.2 Inspection and Testing of Residual Current Devices Form No.: KU – FD – EHS – FOR – 122.

10.3 Electrical Equipment Inspection Checklist No.: KU – FD – EHS – CHE – 26.

10.4 Electrical Record of Maintenance Form No.: KU – FD – EHS – FOR – 123.

10.5 Failed Equipment in Building Form No.: KU – FD – EHS – FOR – 124.

10.6 Register of Failed Equipment in Building: KU – FD – EHS – REG – 30.

Annexure 1

Inspection and Testing of Electrical Equipment Intervals

جامعة خليفة
Khalifa University

Inspection and Testing of Electrical Equipment Intervals

Type of environment and or equipment	Risk Factors			
	High level of cord flexion AND/OR frequency of connection	Low Level of cord flexion AND/OR frequency of connection	Wet or corrosive area or uses water or a corrosive substance in its operation	No cord flexion, no frequency of connection, fixed or stationary equipment
Workshops, places of work or repair, manufacturing, assembly, maintenance or fabrication.	6 months	12 months	6 months	Not Required
Laboratories, health care & educational establishments.	12 months	2 years	12 months	Not Required
Office environment.	12 months	5 years	12 months	Not Required
Hire equipment.	Visually inspect before each hire. Test every 3 months.			
Repaired/serviced/second hand equipment.	After repair or service which could affect electrical safety, or on re-introduction to service			
Equipment used for commercial cleaning.	6 months	12 months	6 months	Not Required
Accommodation houses.	12 months	2 years	12 months	Not Required
Theatre environment	6 months	5 years	12 months	Not Required

MANAGEMENT SYSTEM MANUAL	Inspection and Testing of Electrical Equipment Intervals Form		
Document No: KU – FD – EHS – FOR – 121	Issue no: 02	Issue Date: 07.06.2018	Page: 1 of 1

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Annexure 2

KU Inspection and Testing of Residual Current Devices

جامعة خليفة
Khalifa University

KU Inspection and Testing of Residual Current Devices

Type of environment and or equipment	Push button Test by User		Operating Time RCD tester	
	Portable	Fixed	Portable	Fixed
Workshops, places of work or repair, manufacturing, assembly, maintenance or fabrication.	Daily, or before every use, whichever is longer	6 months	12 months	12 months
Laboratories, health care & educational establishments.	3 months, or before every use, whichever is the longer	6 months	2 years	2 years
Office environment.	3 months	6 months	2 years	2 years
Hire equipment.	Prior to each hire. Test monthly	NA	2 years	2 years
Repaired/serviced/second hand equipment.	After repair or service which could affect electrical health and safety, or on re-introduction to service			
Equipment used for commercial cleaning.	3 months	NA	2 years	NA
Accommodation houses.	6 months	6 months	2 years	2 years
Theatre environment	6 months	6 months	5 years	5 years

MANAGEMENT SYSTEM MANUAL	KU Inspection and Testing of Residual Current Devices Form		
Document No: KU – FD – EHS – FOR – 122	Issue no: 02	Issue Date: 07.06.2018	Page: 1 of 1

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Record of Reviews & Revisions

REV.	DATE ISSUED	DESCRIPTION OF REVISION		REVIEWED BY	AUTHORIZED BY
		PAGE NO	COMMENT		
01	07/06/2018	All	Change of logo	Roberto M.	Belal I.
		Throughout	Change from KUSTAR to KU	Roberto M.	Belal I.
		Throughout	Change the reference from AD EHSMS to OSHAD SF	Roberto M.	Belal I.
		02	Prepared by Reviewed / Updated by, Endorsed for Approval by, and Approval & Issue by	Roberto M.	Belal I.
		25	Change the reference from AD EHS Center to OSHAD	Roberto M.	Belal I.
		25	Reference Standard from ISO 14001:2004 to ISO 14001:2015	Roberto M.	Belal I.
		25	Reference Standard from OHSAS 18001 2007 to ISO 45001:2018	Roberto M.	Belal I.

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